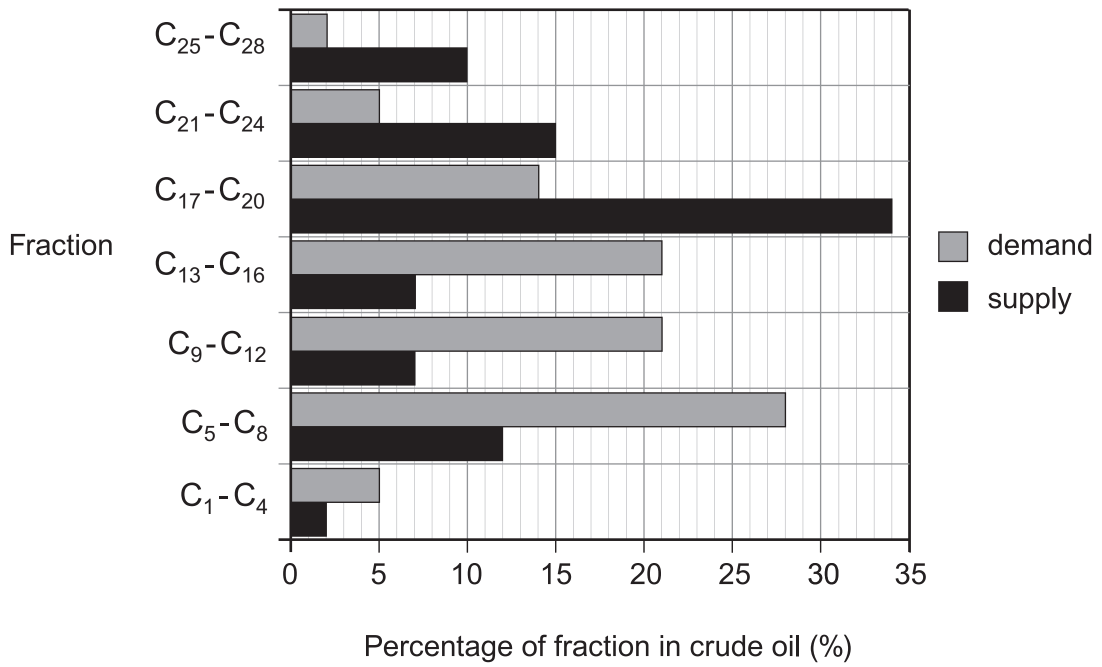
Examiner only

3. *(a)* The bar chart shows the relative *supply* and *demand* for some fractions obtained from crude oil.



(i) Use the bar chart to describe how the **difference** between *supply* and *demand* of the fractions changes as chain length increases. [2]

....

....

....

....

....

Examiner only

(ii) Oil companies have solved the problem of the over-supply of some fractions by using a process called cracking.

$C_{14}H_{30}$ can be cracked forming hexene, ethene and hydrocarbon **A**.

I. Complete the equation for the cracking of $C_{14}H_{30}$. [1]

$$C_{14}H_{30} \longrightarrow C_6H_{12} + \dots\dots\dots + 2C_2H_4$$

hexene hydrocarbon **A** ethene

II. Name hydrocarbon **A**. .... [1]

III. State why ethene is considered an important raw material. [1]

....

....

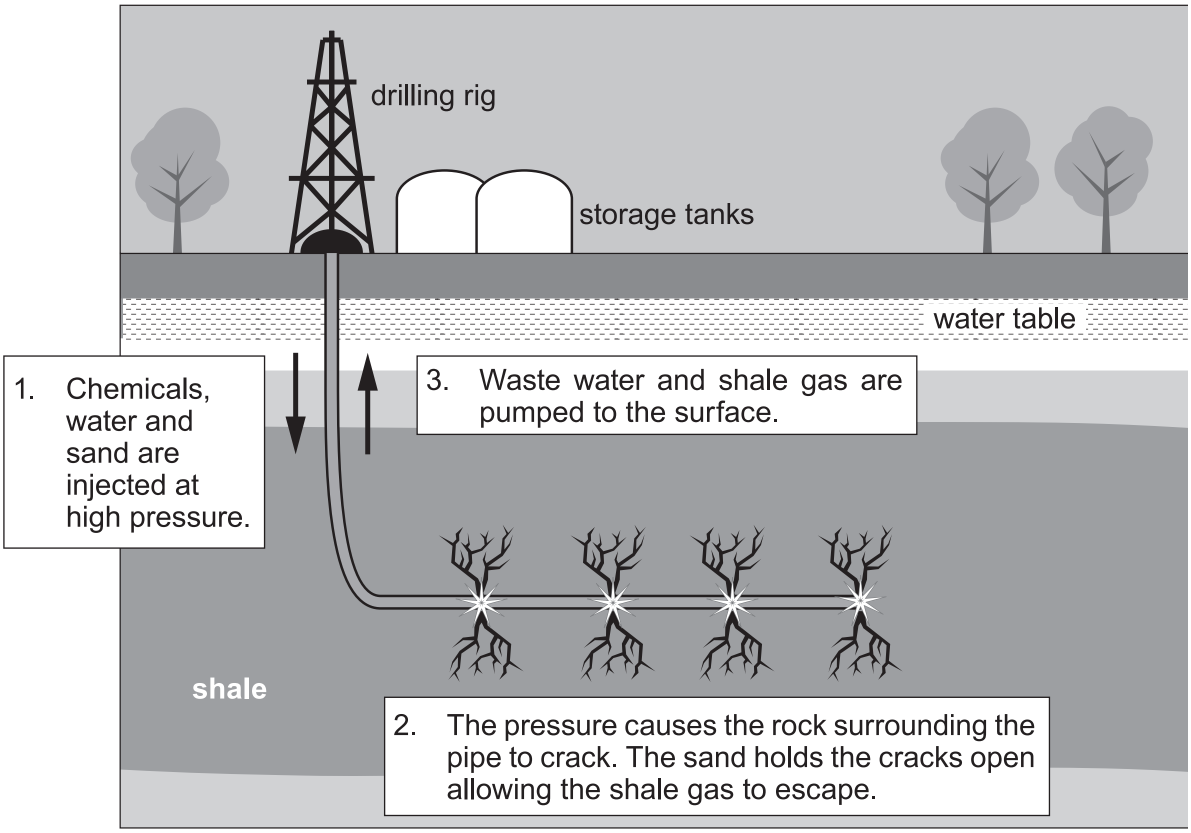
(iii) One hydrocarbon found in the C_1-C_4 fraction is propane. Propane burns in air forming carbon dioxide and water.

Balance the symbol equation that represents this reaction. [1]

$$C_3H_8 + \boxed{}\,O_2 \longrightarrow \boxed{}\,CO_2 + \boxed{}\,H_2O$$

3410UB01 09

(b) Shale gas is natural gas trapped in rocks deep underground. Like oil and coal, shale gas has, essentially, formed from the remains of plants, animals and micro-organisms that lived millions of years ago. It can be extracted using a process known as hydraulic fracturing – or "fracking" – which involves drilling long horizontal wells into the rocks more than a kilometre below the surface. Massive quantities of water, sand and chemicals are pumped into the wells at high pressure. This opens up cracks in the shale, which are held open by the sand, enabling the trapped gas to escape to the surface for collection.



Supporters of fracking argue that extracting shale gas deposits will help keep energy affordable and cut consumption of dirtier coal. But opponents claim fracking is dangerous and polluting, and that tapping into extra shale gas supplies will increase carbon dioxide emissions. The main controversy surrounding shale gas is the potential of fracking to contaminate drinking water supplies with shale gas or drilling chemicals. Other issues include the huge quantities of water and chemicals used in the extraction process, the waste water generated and possible earthquake tremors.

Examiner only

(i) Put a tick (✓) in the box next to the statement that identifies the substance(s) recovered during the fracking process. [1]

only shale gas	☐
shale gas and contaminated water	☐
shale gas and sand	☐
shale gas, sand and toxic chemicals	☐

(ii) Put a tick (✓) in the box next to the statement which is true. [1]

burning shale gas does not cause global warming	☐
chemicals used in fracking are contaminating our drinking water	☐
fracking produces vast quantities of contaminated water	☐
shale gas is a renewable energy source	☐
shale gas is cheaper than other fossil fuels	☐

3410UB01
11

8